INCH-POUND
MIL-DTL-32699
w/AMENDMENT 1
5 March 2025
SUPERSEDING
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15 February 2022

DETAIL SPECIFICATION

CLOTH, 70 DENIER NYLON RIPSTOP

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a 70 denier (70D) nylon ripstop. The fabric is available in one type.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-STD-3064 - Evaluation of Quality of Textile Materials

(Copies of this document are available online at https://quicksearch.dla.mil/.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

U.S. ARMY DEVCOM SOLDIER CENTER

Fabric Defect Replica Scales

(For information on obtaining copies of the Fabric Defect Replica Scales, please contact U.S. Army DEVCOM Soldier Center, ATTN: FCDD-SCP-WC, 10 General Greene Avenue, Natick, MA 01760-5019.)

Comments, suggestions, or questions on this document should be addressed to Marine Corps Systems Command, 2200 Lester Street, Quantico, VA 22134 ATTN: SEAL-SE-STDS or emailed to USMC_STDZ@usmc.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 8305

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CODE OF FEDERAL REGULATIONS (CFR)

16 CFR 303 - Rules and Regulations Under the Textile Fiber Products Identification Act

Title 40 CFR - Protection of Environment

(Copies of these documents are available online at www.ecfr.gov.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC EP1 - Gray Scale for Color Change

AATCC EP2 - Gray Scale for Staining

AATCC EP9 - Evaluation Procedure for Visual Assessment of Color Difference of Textiles

AATCC TM16.3 - Test Method for Colorfastness to Light: Xenon-Arc

AATCC TM22 - Test Method for Water Repellency: Spray Test

AATCC TM61 - Test Method for Colorfastness to Laundering: Accelerated

DELETED

AATCC TM81 - Test Method for pH of the Water-Extract from Wet Processed Textiles

AATCC TM127 - Test Method for Water Resistance: Hydrostatic Pressure

AATCC TM135 - Dimensional Changes of Fabrics after Home Laundering

(Copies of these documents are available online at www.aatcc.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at www.asq.org.)

DELETED

ASTM INTERNATIONAL

ASTM D1424 - Standard Test Method for Tearing Strength of Fabrics by Falling-Pendulum (Elmendorf-Type) Apparatus

ASTM D3775 - Standard Test Method for End (Warp) and Pick (Filling) Count of Woven Fabrics

ASTM D3776/D3776M - Standard Test Methods for Mass Per Unit Area (Weight) of Fabric

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ASTM D5034 - Standard Test Method for Breaking Strength and Elongation of Textile Fabrics (Grab Test)

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(Copies of these documents are available online at www.astm.org.)

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2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Inspections.

3.1.1 First article. When specified (see 6.2), a sample shall be subjected to a first article inspection (see 6.3) in accordance with 4.2.

3.1.2 Conformance inspection. When specified (see 6.2), a sample shall be subjected to a conformance inspection in accordance with 4.3.

3.2 Standard sample. Unless otherwise specified (see 6.2), the finished fabric shall match the standard sample (see 6.4) for shade and appearance, and shall be equal to or better than the standard sample with respect to all other characteristics for which the standard sample is referenced.

3.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Basic material. The basic material for the fabric shall be constructed from a 70D nylon with a water repellant finish in a ripstop weave.

3.5 Color. The basic color shall be Coyote 498.

3.5.1 Visual shade matching. The color and appearance of the cloth specified in 3.4 and 3.5 shall match the standard sample when tested as specified in 4.4.7.

3.5.2 Colorfastness. The finished exterior of the cloth shall show colorfastness to light and laundering. Unless otherwise specified in the contract or procurement documents (see 6.2), the finished cloth referenced in 3.4 shall conform to the colorfastness requirements listed below in table I when tested as specified in 4.4.6.1.

TABLE I. Colorfastness requirements (all classes).

Colors Evaluation	**Laundering - Color Change and Staining (3 Cycles) (Minimum Rating)**	**Light (After 40 AFU [1/] or 170 kJ/(m^{2}nm) [2/] at 420 nm) (Minimum Rating)**
All colors	3.5	3.5
FOOTNOTES: 1/ AATCC Fading Units. 2/ Kilojoules (kJ) per square meter per nanometer.		

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3.5.3 Spectral reflectance. The finished outershell cloth shall meet the spectral reflectance values (in percent) of table II for the visible/near infrared wavelength range, 600 to 860 nanometers, for the colors specified in 3.5, as applicable, when tested as specified in 4.4.9.

TABLE II. Spectral reflectance.

Wavelengths	Coyote 498	
Nanometers	Min.	Max.
600	8	20
620	8	20
640	8	22
660	8	24
680	12	24
700	12	34
720	16	42
740	22	46
760	30	50
780	34	54
800	36	56
820	38	58
840	38	58
860	40	60

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3.6 Physical requirements. The fabric shall be a woven ripstop made from 70D nylon and shall meet the requirements stated in table III when tested as specified in 4.4.

TABLE III. Physical requirements.

Characteristic	Requirement
Weight, ounces per square yard	2.0 – 2.5
Yarns per inch (minimum)	
Warp	100
Fill	85
Breaking strength, pounds (minimum)	
Warp	125
Fill	110
Elongation, percent (minimum)	
Warp	40
Fill	45
Tear strength, pounds (minimum)	
Warp	3.5
Fill	4.5
DELETED	
Drying time, minutes (maximum)	30
Spray rating (minimum)	100, 100, 100
Dimensional stability, percent (maximum)	
Warp	±5
Courses	±5
Hydrostatic, centimeters (minimum)	30

3.7 Finish. The nylon ripstop fabric shall have a durable water-repellent finish applied to one side only.

3.8 pH. The pH of the water extract of the finished cloth shall be no less than 5.0 nor more than 8.5 when tested as specified in 4.4.6.1.

3.9 Toxicity. The finished fabric shall not present a health hazard and shall show compatibility with prolonged, direct skin contact when tested as specified in 4.4.8. Chemicals recognized by the Environmental Protection Agency (EPA) as human carcinogens shall not be used.

3.10 Length, width, and put-up.

3.10.1 Length and put-up. Unless otherwise specified (see 6.2), for Government procurements only, the cloth shall be furnished in continuous lengths, each not less than 40 yards, and each length shall be put-up on a roll as specified (see 6.2) when examined as specified in 4.4.4.

3.10.2 Width. Unless otherwise specified (see 6.2), for Government procurements only, the width of the finished cloth shall be as specified (see 6.2) and shall be the minimum acceptable width inclusive of the selvage.

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3.11 Roll identification.

3.11.1 Face identification. The water repellant treated side shall be identified as the face side by stamping that side with the word "FACE" at each end of the roll.

3.11.2 Fiber identification. Each roll shall be labeled or ticketed for fiber content in accordance with 16 CFR 303 when examined as specified in 4.4.

3.12 Workmanship. The finished cloth shall conform to the quality and grade of product established by this specification, be uniform, and be free from defects specified in 4.4.5. The occurrence of defects shall not exceed the established Acceptance Quality Limits (AQL) (see 6.2) and shall not adversely affect the serviceability, appearance, and uniformity of the product.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. Unless otherwise specified in the contract (see 6.2), a first article, submitted in accordance with 3.1.1, shall be examined and tested in accordance with 4.4.1 through 4.4.10 for compliance with the appearance, color, finish, design, workmanship, and testing requirements specified in this document.

4.2.1 First article samples and acceptance criteria. Unless otherwise specified in the contract (see 6.2), a first article sample shall be provided. The sample unit shall be one continuous 5-yard length of full width cloth. The presence of excessive defects or failure of any testing shall be cause for rejection of the first article.

4.3 Conformance inspection. Conformance inspection shall consist of the examinations and tests as specified in 4.4 through 4.4.10. The Government reserves the right to inspect and test all components, materials, and end items to determine conformance to stated requirements.

4.3.1 Conformance inspection samples. Sampling for conformance inspection shall be performed in accordance with ASQ/ANSI Z1.4 and with AQLs as specified in the contract or order (see 6.2). The sample unit shall be one yard of finished cloth, and the lot size shall be expressed in yards.

4.4 Inspections. Unless otherwise specified, excluded, amended, modified or qualified in this specification or applicable procurement documents (see 6.2), all inspections shall be performed in accordance with all the requirements of referenced documents.

4.4.1 Visual inspection. Each roll in the sample shall be examined yard-by-yard on the face side for defects in accordance with MIL-STD-3064, type I. In the event of a conflict between the defects cited in MIL-STD-3064 and those in 4.4.5, this specification takes precedence.

4.4.2 Roll identification and marking examination. During the yard-by-yard examination, each roll in the sample shall be examined for defects as specified in MIL-STD-3064.

4.4.3 Shade variation examination. During the yard-by-yard examination, each roll in the sample shall be examined for shade variation as specified in MIL-STD-3064.

4.4.4 Length examination. For Government procurements only, during the yard-by-yard examination, each roll in the sample shall be examined for length as specified in MIL-STD-3064.

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4.4.5 Classification of defects. The required yardage shall be examined for defects as listed in table IV. The fabric shall be examined on the face side and the back side. The defects found shall be counted individually, regardless of their proximity to each other, except where two or more defects represent a single local condition of the cloth, in which case only one defect shall be counted. A continuous defect shall be counted as one defect for each yard or fraction of a yard in which it occurs. The sample unit shall be 1 linear yard. The lot size shall be expressed in units of 1 yard each. The AQLs shall be as specified in the contract or order (see 6.2). The sample size and number of rolls shall be in accordance with table V.

TABLE IV. Fabric defects.

Defect Description	Points
Any hole, cut, tear, or abrasion mark.	4
Edge ravels when pulled outward.	4
Broken or missing yarn.	4
Filling or warp threads missing, heavy filling or warp, kinky filling or warp, or floats greater than 9 inches.	4
Any spot, stain, or streak more than 1 inch in combined directions.	4
Color not a good match to reference.	4
Any area of poor dye penetration or a streak or shade bar that is 9 inches or less. NOTE: Defective shade areas that are greater than 9 inches constitute a failure.	4
Width not as specified within a tolerance of less than 1/2 inch.	4
Objectionable odor.	4
Slack or tight selvages.	4
Any embedded foreign matter or uncleanness.	4
Edges rolled, folded, creased, scalloped, or corded or baggy, ridgy, or wavy cloth.	4

TABLE V. Physical requirements verification sampling.

Lot Size	Sample Size (# of Sample Units)
800 or less	2
801 through 22,000	3
22,001 and over	5

4.4.6 Physical requirements verification. Unless otherwise specified in the contract (see 6.2), the cloth shall be examined for conformance to the physical requirements listed in table III. The methods of testing shall be as specified in table VI.

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TABLE VI. Physical requirements verification.

Characteristic	Requirement	Verification
Weight, ounces per square yard	3.6	ASTM D3776/D3776M
Yarns per inch		
Warp	3.6	ASTM D3775
Fill	3.6	ASTM D3775
Breaking strength, pounds		
Warp	3.6	ASTM D5034
Fill	3.6	ASTM D5034
Elongation, percent		
Warp	3.6	ASTM D5034
Fill	3.6	ASTM D5034
Tear strength, pounds		
Warp	3.6	ASTM D1424
Fill	3.6	ASTM D1424
DELETED		
Drying time, minutes	3.6	In-house method (see 4.4.10)
Spray rating	3.6	AATCC TM22
Dimensional stability, percent	3.6	
Warp	3.6	AATCC TM135, 3, III, Aiii
Courses	3.6	AATCC TM135, 3, III, Aiii
Hydrostatic, centimeters	3.6	AATCC TM127
pH	3.8	AATCC TM81
Visual shade matching	3.5.1	AATCC EP9, option C
Colorfastness evaluation		
Laundering (after three cycles)	3.5.2	AATCC TM61 test 3A 1/
Light	3.5.2	AATCC TM16.3, option 3 2/

FOOTNOTES:

1/ Rated using AATCC EP1 and AATCC EP2.

2/ Rated using AATCC EP1.

4.4.6.1 Sampling and acceptance criteria for verification of physical requirements. Sampling shall be in accordance with table V. The sample unit shall be 5 continuous yards, full-width, of the finished cloth per roll of cloth in the sample. The lot shall be unacceptable if one or more sample units fail to meet any test requirements specified in this document.

4.4.7 Visual shade matching. The color and appearance of the 70D nylon ripstop shall match the standard sample under sources simulating artificial daylight D65 illuminant (see 6.5.1) with a color temperature of 6500±200 Kelvin and illumination of 100±20 foot candles. Under these conditions, the color and appearance of the final fabric shall also be a good match to the standard sample under incandescent A illuminant with a color temperature of 2856±200 Kelvin.

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4.4.8 Toxicity test. When specified (see 6.2), an acute dermal irritation study and a skin sensitization study shall be conducted. When the results of these studies indicate the fabric may be a sensitizer or irritant, testing shall be conducted in accordance with Title 40 CFR (see 6.7). If the toxicity requirement (see 3.9) can be demonstrated with historical use data on the finishing treatments used, toxicity testing may not be required (see 6.2).

4.4.9 Spectral reflectance test. Spectral reflectance data shall be obtained from 600 to 860 nanometers at 20-nanometer intervals on a spectrophotometer relative to the polytetrafluoroethylene (PTFE) family of compounds, the preferred white standard. Other white reference materials may be used provided they are calibrated to absolute white or vitrolite tiles. The spectral band width shall be less than 20 nanometers at 860 nanometers. Reflectance measurements shall be made by either the monochromatic or polychromatic mode of operation. When the polychromatic mode of operation is used, the spectrophotometer shall operate with the specimen diffusely illuminated with the full emission of a continuous source that simulates either International Commission on Illumination (CIE) source A or CIE source D65. Measurements shall be taken on a minimum of two different areas and the data averaged. The measured areas should be at least 6 inches away from the selvage. The specimen shall be measured as a single layer backed with three layers of the same fabric and shade. The specimen shall be viewed at an angle no greater than 10 degrees from normal with the specular component included. Specimens shall be oriented in different directions during testing. When possible, the specimens tested shall not contain the same warp or filling yarns or wales or courses when presented to the sample port. Camouflage materials should be measured with the appropriate aperture size to ensure that only one color is measured at a time. The largest aperture possible should be used. Photometric accuracy of the spectrophotometer shall be within one percent, and wavelength accuracy shall be within 2 nanometers. Any color having spectral reflectance values falling outside the limits at four or more of the wavelengths specified shall be considered a test failure.

4.4.10 Drying time. The drying time test shall be run in standard textile conditions (65±5 percent relative humidity and 70±4 degrees Fahrenheit) on three specimens. The two-inch by two-inch (2" x 2") fabric samples and blotting paper shall be conditioned at standard textile conditions for a minimum of 4 hours. The weight of the conditioned specimen shall be measured using a laboratory balance, accurate to 0.01 gram. A wire mesh kitchen/bathroom sink strainer may be used by placing it on the weight pan of the lab balance in an inverted manner and taring its weight before the measurement of any specimen weight. After weighing the samples, 100 milliliters of distilled water shall be placed into a 250-milliliter glass beaker. The specimen shall be completely submerged (to ensure complete wetting) in the beaker of water for 30 minutes. The specimen shall then be removed and placed between two pieces of unused blotting paper. The blotting paper-enclosed specimen shall be passed through a motor-driven wringer (see 6.6.2) with a dead weight load of 27.2±0.5 kilograms (60±1 pounds) and speed of 1 inch per second. The specimen shall immediately be placed on the balance with the top door of the balance open and the side doors closed. The wet weight shall be recorded to the nearest 0.01 gram. The weight shall be manually monitored at set intervals until dry; alternatively, an automated balance with the capability to weigh the specimen until dry may be used. The degree of dryness shall be determined by the user. The time to dry (t) shall be recorded. This process should be repeated for the remaining specimens.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification covers 70D nylon ripstop used in lining material for handwear by the United States Marine Corps.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. The specific issue of individual documents referenced (see 2.2 and 2.3).
c. Color required.
d. When first article sample is required (see 3.1.1, 4.2, and 6.3).
e. National stock number.
f. When conformance inspection is required (see 3.1.2).
g. Applicable Government standard samples, patterns, and drawings, including revisions (see 3.2).
h. Requirements for colorfastness (see 3.5.2).
i. When toxicity testing is required (see 3.9 and 4.4.8).
j. Length, width, and put-up required (3.10.1 and 3.10.2).
k. AQLs required (see 3.12, 4.3.1, and 4.4.5).
l. Number of first article inspection samples (see 4.2.1).
m. Requirements and conditions for inspections (see 4.4).
n. Requirement for verification of physical requirements (see 4.4.6).
o. DELETED
p. Packaging requirements (see 5.1).

6.3 First article. When first article inspection (see 3.1.1) is required (see 6.2), the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory, and the number of items to be tested as specified in 4.2. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Standard sample. For standard samples, address the procuring activity issuing the invitation for bids or request for proposal.

6.5 Visual shade matching. In 2019, option A of AATCC EP9 was changed to option C. NOTE: In case of confusion, the viewing geometry should be "The specimen plane and illumination source will be parallel to each other and aligned so that the light flux is incident at the center of the specimen plane, which is set at a 35 (±5°) angle relative to the horizontal. The observer will view the specimens at a 90° angle, relative to the plane of the specimens".

6.5.1 Use of D75 illuminant. The use of D75 illuminant with a color temperature of 7500±200 Kelvin and illumination of 100±20 foot candles in lieu of the specified D65 illuminant (see 4.4.7) is permitted.

6.6 Sources of supply.

6.6.1 Basic fabric. A known source of fabric meeting the requirements of this specification is Brookwood Companies, Inc. (www.brookwoodcos.com or 485 Madison Avenue, Suite 500, New York, NY 10022).

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6.6.2 Powered wringer. A known source of a powered wringer meeting the requirements of this specification is SDL Atlas Model D394 Laboratory Wringer available from SDL Atlas (https://sdlatlas.com, 803-329-2110, or 3934 Airway Drive, Rock Hill, SC 29732).

6.7 Toxicity testing. OECD Test No. 406: Skin Sensitization and OECD Test No. 404: Acute Dermal Irritation/Corrosion adequately have been suggested to assess this requirement, but other methods may be used.

6.8 Subject term (key word) listing.

Lining

Water resistant

6.9 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

CONCLUDING MATERIAL

Custodians:
Army – GL
Navy – MC
Air Force – 11

Preparing activity:
Navy − MC
(Project 8305-2025-002)

Review activities:
Army – AV, CR, IH, MI
Navy – AS, CG1, NU
Air Force – 03
DLA – CT

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